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About

Dr. Xin Huang is an Associate Professor at the Department of Computer Science, Hong Kong Baptist University. He received the Ph.D. degree from the Department of Systems Engineering and Engineering Management at the Chinese University of Hong Kong in 2014. His research interests include graph data management, graph mining and visualization, social network analysis, and privacy-aware computing. He has published in refereed conferences and journals including SIGMOD, PVLDB, ICDE, WWW, DAC, CIKM, AAAI, IJCAI, TKDE, TKDD, TMC, TNNLS, and VLDBJ. His scholarly works win the Best Paper Award at the 20th International Conference on Web Information Systems Engineering (WISE 2019), the Best Student Paper Award at the 11th IEEE International Conference on Semantic Computing (ICSC 2017), the Best Paper Nomination at the 29th ACM International Conference on Information and Knowledge Management (CIKM 2020), as well as the Best Student Paper Award at the 32nd ACM/IEEE International Workshop on Logic & Synthesis (IWLS 2023). He is a recipient of the RGC Early Career Award (2020), the HKBU President's Award for Outstanding Performance as Young Researcher (2021), and the Departmental Best Research Award (2020) and Best Teaching Award (2023).

Research Interests

- Data Management
- Big Graph Mining & Visualization
- Social Network Analysis
- Privacy-Aware Computing

Selected Publications

- L. Sun, X. Huang, Z. Wu, and J. Xu, "Efficient Cross-layer Community Search in Large Multilayer Graphs", IEEE International Conference on Data Engineering

(**ICDE'24**), 2959-2971, 2024.

- Z. Sun, X. Huang, C. Piao, C. Long, and J. Xu, "Adaptive Truss Maximization on Large Graphs: A Minimum Cut Approach", IEEE International Conference on Data Engineering (**ICDE'24**), 3270-3282, 2024.
- Z. Sun, X. Huang, Q. Liu, and J. Xu, "Efficient Star-based Truss Maintenance on Dynamic Graphs", Proceedings of the ACM on Management of Data (**SIGMOD'23**), 1(2): 133: 1-133:26, 2023.
- X. Sun, X. Huang, and D. Jin, "Fast Algorithms for Core Maximization on Large Graphs", Proceedings of the VLDB Endowment (**PVLDB'22**), 15(8): 1350-1362, 2022.
- J. Huang, X. Huang, and J. Xu, "Truss-based Structural Diversity Search in Large Graphs", IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 34(8):4037-4051, 2022.
- L. Sun, X. Huang, R. Li, B. Choi, and J. Xu, "Index-Based Intimate-Core Community Search in Large Weighted Graphs", IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 34(9): 4313-4327, 2022.
- X. Zhu, X. Huang, B. Choi, J. Jiang, Z. Zou, and J. Xu, "Budget Constrained Interactive Search for Multiple Targets", Proceedings of the VLDB Endowment (**PVLDB'21**), 14(6): 890-902, 2021.
- Z. Dong, X. Huang, G. Yuan, H. Zhu, and H. Xiong, "Butterfly-Core Community Search over Labeled Graphs", Proceedings of the VLDB Endowment (**PVLDB'21**), 14(11): 2006-2018, 2021.
- Z. Sun, X. Huang, J. Xu, and F. Bonchi, "Efficient Probabilistic Truss Indexing on Uncertain Graphs", The Web Conference (**WWW'21**), 354-366, 2021.
- Y. Jiang, X. Huang, and H. Cheng, "I/O Efficient K-Truss Community Search in Massive Graphs", The VLDB Journal (**VLDBJ**), 30(5): 713-738, 2021.
- X. Zhu, X. Huang, B. Choi, and J. Xu, "Top-k Graph Summarization on Hierarchical DAGs", ACM International Conference on Information and Knowledge Management (**CIKM'20**), 1903-1912, 2020.
- X. Huang and L. V.S. Lakshmanan, "Attribute-Driven Community Search", Proceedings of the VLDB Endowment (**PVLDB'17**), 10(9): 949-960, 2017.
- X. Huang, W. Lu, and L. V.S. Lakshmanan, "Truss Decomposition of Probabilistic Graphs: Semantics and Algorithms", ACM SIGMOD International Conference on Management of Data (**SIGMOD'16**), 77-90, 2016.
- X. Huang, L. V.S. Lakshmanan, J. X. Yu, and H. Cheng, "Approximate Closest Community Search in Networks", Proceedings of the VLDB Endowment (**PVLDB'15**), 9(4): 276-287, 2015.
- X. Huang, H. Cheng, L. Qin, W. Tian, and J. X. Yu, "Querying K-Truss Community in Large and Dynamic Graphs", ACM SIGMOD International Conference on

Management of Data (**SIGMOD'14**), 1311-1322, 2014.